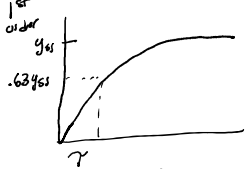
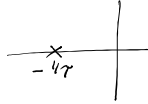


# Transient Specifications: Quantify Performance

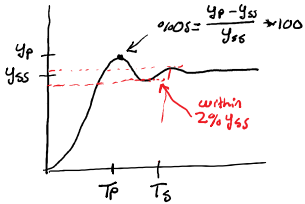


$$G(s) = \frac{K}{\tau s + 1}$$

$$G(s) = \frac{K/\tau}{s + 1/\tau}$$

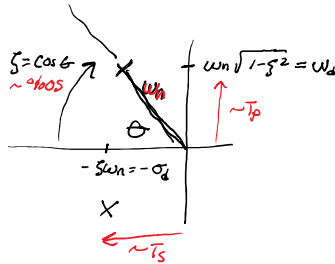


Response  $\longleftrightarrow$  Poles



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} K$$

$\omega_n$  = natural frequency  
 $\zeta$  = damping Ratio



$$T_p = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\sigma_d}$$

$$\%OS = F(\zeta)$$

## Control Design Based on Pole Location:

DC Motor Position Control:

(stage)

(LAB 5)

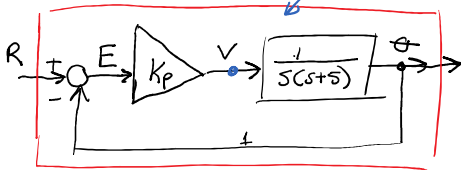
(LAB 2)

$$G_w(s) = \frac{K}{\tau s + 1} = \frac{\text{Space}}{\text{Voltage}}$$

Example:  $G_\theta = \frac{1}{s(s+\bar{s})}$

Control Law  
 $V = K_p E = K_p(R - \theta)$

$$G_\theta(s) = \frac{1}{s} \frac{K}{\tau s + 1} = \frac{\text{Position}}{\text{Voltage}}$$



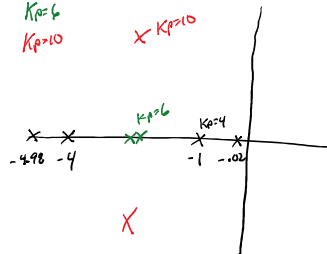
$$G_{OL} = \frac{K_p}{s(s+\bar{s})} \frac{s(s+\bar{s})}{1 + \frac{K_p}{s(s+\bar{s})}}$$

$$G_{CL} = \frac{K_p}{s(s+\bar{s}) + K_p} = \frac{K_p}{s^2 + \bar{s}s + K_p}$$

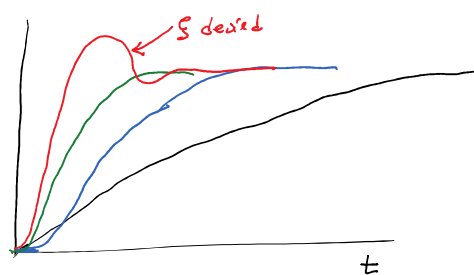
Poles:  $s^2 + \bar{s}s + K_p = 0$  O.E.

Pick  $K_p$  to get Poles (Response) desired

- $K_p = 1$
- $K_p = 4$
- $K_p = 6$
- $K_p = 10$



Root Locus diagram



## Design $K_p$ with model:

Example  $G_{CL} = \frac{K_p}{s^2 + \bar{s}s + K_p}$

desired  $G_{desired} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

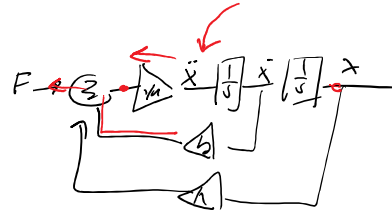
equating coefficients

design for 2% overshoot

$$\zeta = \frac{-\ln(\frac{2}{100})}{\sqrt{\pi^2 + \ln^2(\frac{2}{100})}} = 0.78$$

$\Rightarrow$  2 equations  
2 unknowns  $\omega_n, K_p$   
\* Solve \*

equation  $\longleftrightarrow$  block diagram



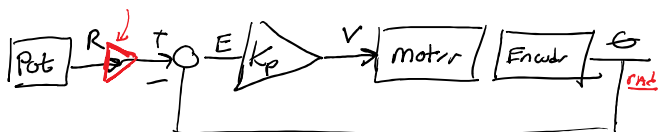
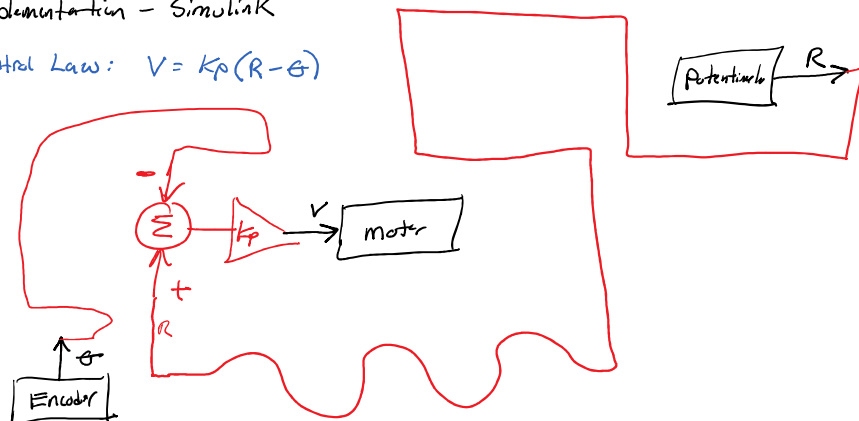
$$\ddot{x} = \frac{1}{m}(F - bx - Kx)$$

$$K_p = \omega_n^2 \quad 5 = 25\omega_n \quad \Rightarrow K_p = \left(\frac{5}{25}\right)^2 = \left(\frac{5}{25 \cdot 70}\right)^2 = 10.2$$

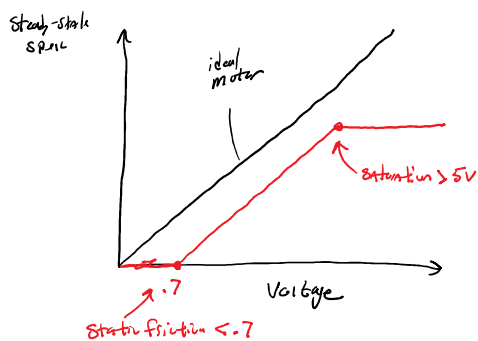
1st Page

Implementation - Simulink

Control Law:  $V = K_p(R - \theta)$



Implementation Constraints



How do these affect the system / control system?

• Saturation

Car Example:

Control Law:

$$\text{Pedal} = K_p E = K_p (\text{desired speed} - \text{speed})$$

$$K_p = 20$$

$$\text{Pedal} = 20(10) = 200$$

↑  
not possible

Saturation will limit  $K_p$

• Static Friction

Control Law:

$$V = K_p E = K_p (\theta_d - \theta)$$

when this  $< 0.7$  Volts  $\rightarrow$  motor stops

$K_p$  too small  $E$  too small  $< 0.7$



Static friction & Saturation — use control law

Unit Conversions :

Experiment - Known Value:

move minute hand  $90^\circ$

Data : Result from Experiment

minutes changed 15 minutes

